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Does Online Social Media Lead to Social Connection or Social Disconnection?

Jean M. Twenge, San Diego State University[1]

Abstract

Today's young generation (often called *Millennials, GenY,* or *Generation Me*) are the first to grow up with the Internet and social networking websites. Have these experiences led to more and better social connections, or fewer and atrophied ones? Social media use may lead to online political action such as signing an e-mail petition but does not appear to lead to other engagement in public life (such as writing to public officials or having more knowledge about politics). Youth who spend more time online are more likely to fight and carry weapons and are less likely to help others, although the direction of causation is unclear. Social media build shallow, "weak" ties, increase self-focus (including narcissism), and may lead to mental health issues for some individuals. Over the time social media became popular, young people's empathy for others, civic engagement, and political involvement declined.

Just twenty years ago, cell phones were rare, the Internet was virtually unknown, and few people used e-mail. Now college students not only use all of this technology, but keep it at their fingertips on their smartphones. A Kaiser Family Foundation study found that teens were online or on cell phones for seven and a half hours every day (Rideout, Foehr, & Roberts, 2010). The rapid pace of change in such a short period has led many to wonder how it has affected young people and how they differ from earlier generations as a result.

Of course, older generations have "always" worried about how technology impacts youth. For example, many in the 1960s and 1970s expressed concerns about excessive television viewing. "But," the argument often goes, "my generation watched lots of TV and we turned out fine." But did we? In *Bowling Alone*, Robert Putnam (2000) traced the decline in community groups and organizations to increased TV viewing. In this case, technology created false social relationships that replaced real social relationships in the community.

At first glance, any such worry about cell phones and the Internet seems misplaced—after all, these technologies connect people rather than, like television, isolate them in their homes. However, some have

[1] Jean M. Twenge is professor of psychology at San Diego State University. She is the author of more than 100 scientific publications and the books *Generation Me* and *The Narcissism Epidemic* (with W. Keith Campbell). E-mail: jtwenge@mail.sdsu.edu.

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argued that social media and other modern technologies isolate people rather than connect them. As they interact through screens, they may lose the ability to relate to others in person (e.g., Rosen, 2012; Turkle, 2012).

Empirical research on the effects of social media and the Internet is still in its infancy. In this article, I will review the evidence—both direct and indirect—on whether this new technology, particularly social media, has led to an increase or decrease in caring, community, social action, and robust mental health among high school and college students—Generation Me (also known as GenMe; Twenge, 2006). In other words: Is social media use creating or undermining beneficial social connections?

These technologies are so new that research is only now beginning to explore its effects on individuals, much less their connections to others and their communities. The research that exists, however, is fairly consistent: It suggests that social media lead to more connections, but shallower connections (sometimes known as "weak ties"), and that they increase low-effort online engagement such as online petitions but not other participation in or knowledge of civic life.

Internet use may also lead to problematic behaviors among some individuals, including antisocial behavior and mental health issues. Most of these studies are correlational, so the causation is not clear: Internet use may lead to problematic behaviors, problematic behaviors may lead to more Internet use, or the same types of people might engage in both problematic behaviors and more Internet use, making it appear that they cause each other when each might instead indicate another underlying problem (such as an unstable family environment).

Social Media, Social Capital, and Political and Civic Participation

Social media use has become pervasive. Facebook had 850 million active users as of 2012. As of 2010, Americans spent 23% of their Internet time on social networking sites (Nielsen, 2010). Among one sample of college students, 78% logged on to Facebook at least once a day, and 21% checked in more than five times a day (Kittinger, Correla, & Irons, 2012). College students primarily use Facebook to keep in touch with friends they do not see very often and to fill up time and not be bored. Ten percent or less use it to "voice . . . opinions on various topics" or "to share . . . favorite music/video clips." About 30% say they use it "to meet new people and to make new friends" (Subrahmanyam, Reich, Waechter, & Espinoza, 2008).

Initial research conducted at Michigan State University in 2006 suggested that college students who spent more time on Facebook experienced higher levels of social capital—particularly for weak ties (broader, but shallower relationships) and somewhat for strong ties (less broad but deeper relationships; Ellison, Steinfield, & Lampe, 2007). A follow-up study found that Facebook use predicted social capital rather than social capital predicting Facebook use, suggesting social networking sites build social capital (Steinfield, Ellison, & Lampe, 2008). However, both of these studies examined only perceived social capital (such as "I feel I am part of the MSU community"; "I am interested in what goes on at MSU") rather than actual help received or activities commenced.

In a later study, college students at the University at Buffalo asked 12 of their Facebook friends for minimal help with a class project via an e-mail asking them to click on a link and label a few images with words. Only 21% of requests were answered, and the students who perceived greater social capital did not necessarily receive more help (Stefanone, Kwon, & Lackaff, 2012). This study suggests that Facebook may improve perceived social capital but not actual social capital.

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A recent study specifically examined whether college students' online engagement in a social issue (the 2011 Wisconsin public employee union bill) translated to participation in offline protests. Students who used Facebook to express their opinion were more likely to participate in offline protests, but students who used Facebook to gather information were not any more likely to participate offline. Thus students who were already interested in the issue used Facebook to tell others about their interest, but learning about the issue on Facebook did not produce offline action (Macafee & De Simone, 2012). However, another study found that information seeking on Facebook was correlated with political participation (Park, Kee, & Valenzuela, 2009).

In a study conducted in October 2008, college students who spent the most time and had the most friends on Facebook were more likely to engage in political activity on Facebook but significantly *less* likely to participate in politics outside of the site (such as watching a political debate, signing a petition, or writing to a government official). Overall, students were most likely to engage in "lightweight" political participation on Facebook—what has sometimes been called *slacktivism* (interest or engagement with social issues that requires little effort). Most students reported that they believed that using Facebook to persuade others to vote, either in general or for a specific candidate, was not appropriate. The researchers noted that "personal expression is seen as more appropriate than calls to action" (Vitak, Zube, Smock, Carr, Ellison, & Lampe, 2011, p. 112).

Another survey of college students in December 2007 found similar results: Social media users were no more likely to participate in politics (e.g., writing to a politician, boycotting a product), with the exception of online participation such as signing a web petition or forwarding a political e-mail or link. They were also no more likely to have knowledge about politics (e.g., to know the name of the chief justice of the Supreme Court or which party had a majority in Congress) and only slightly more likely to have specific knowledge about that year's presidential primary candidates. They were also no more likely to have voted in the 2006 elections or to say they were likely to vote in the 2008 elections. In contrast, students who watched TV news programs were significantly better informed on political issues and were more likely to vote (Baumgartner & Morris, 2010).

Research on adults has also produced mixed results. A survey of Texas adults found that reliance on social networking sites was correlated with increased civic participation (such as volunteering with a community group or participating in a protest) but not with political participation (signing a petition or working for a political campaign; Zhang, Johnson, Seltzer, & Bichard, 2010). Those who frequently use the Internet, use cell phones, and share photos online also have a greater number of "weak ties" offline. However, those who use social networking services also know fewer of their neighbors (Hampton, Lee, & Her, 2011). As with all correlational studies, these studies cannot determine if offline involvement leads to greater Internet use, if Internet use leads to offline involvement, or (perhaps most likely) whether the same types of people are connected both online and offline. Both studies controlled for some of these confounding variables (most notably, age), but not others (e.g., personality traits such as extraversion or a general tendency to connect with others).

The Effect of Social Media on Antisocial Behavior and Mental Health Issues

Two studies of adolescents suggest that social media and Internet use may lead to more negative behaviors toward others. A study of 12- to 18-year-olds in Hong Kong found that those who used the Internet more

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hours a week were more likely to engage in antisocial and delinquent behavior and less likely to help others (Ma, Li, & Pow, 2011). In a sample of high school students in Connecticut, those with problematic levels of Internet use were more likely to get into serious fights or carry a weapon (Liu, Desai, Krishnan-Sarin, Cavallo, & Potenza, 2011). Because these studies were correlational, it is not clear which behavior is causing the other, or if the same types of youth are drawn to excessive Internet use and antisocial behavior.

Internet use may disinhibit aggressive behavior because users are often anonymous (the technical term is *deindividuation*). In a classic study, women who wore hoods over their heads, and were thus anonymous, administered higher levels of shock to a victim than women who wore their regular clothing and were identifiable (Zimbardo, 1970). On many message boards and comment sections online, users can hide behind their anonymity and, as a result, are more aggressive than they might be face-to-face. Some researchers have labeled this the *online disinhibition effect*—the tendency of people to be less inhibited online, in both good but mostly negative ways (Suler, 2004). And although Facebook is not an anonymous site, it may still remove inhibitions against aggression that would normally be in place in a face-to-face interaction (Turkle, 2012; Wilcox & Stephen, in press).

Cyberbullying, in which users harass and act aggressively toward a victim online, has become common (e.g., Mishna, Cook, Gadallo, Daciuk, & Solomon, 2010). In several high-profile cases, young people have committed suicide after being cyberbullied—for example, Rutgers student Tyler Clementi committed suicide after his roommate live-streamed his gay sexual encounter. In other cases, cyberbullying can include violent threats. In October 2012, film director Michael Gallagher found personal information about him and his girlfriend posted online, along with specific threats of violence against them. One hacker interviewed about the incident was almost nonchalant about it: "I think anybody who's hung around the Internet long enough knows that death threats are thrown around like they're nothing," he said (Taibbi, 2012).

Internet use may also cause or exacerbate mental health issues for some individuals. Many recent studies have examined Internet addiction, a disorder that may be included in the next edition of the *Diagnostic and Statistical Manual of Mental Disorders.* American high school students with problematic levels of Internet use were more likely to make poor grades, smoke cigarettes or marijuana, drink alcohol, or feel sad or hopeless (Liu et al., 2011). Another study found that those with signs of Internet addiction were more likely to be depressed (Morrison & Gore, 2010). As always, Internet addiction could lead to negative behaviors, negative behaviors could lead to Internet addiction, or a third variable (e.g., being prone to addiction overall) could lead to both. Liu et al. (2011) found that about 4% of high school students displayed definite signs of Internet addiction, and Kittinger et al. (2012) found that 16% of college students showed signs of problematic Internet use, including feeling addicted, spending too much time online, or being late to class or other activities.

Even normal amounts of Facebook use may lead to mental health issues, particularly among GenMe users (Rosen, Whaling, Rab, Carrier, & Cheever, 2012). Compared to other generations, GenMe was more likely to report feeling anxious when they could not access Facebook or text messages. Among all generations, Facebook use was correlated with indicators of personality disorders (tendencies and behaviors that impair everyday functioning). Those who used Facebook more often reported more symptoms of bipolar/mania disorders, narcissism, histrionic personality disorder, and antisocial personality disorder. People high in histrionic traits and narcissism also had more friends on Facebook. In a piece of good news, those with more Facebook friends were less likely to be depressed (Rosen et al., 2012).

The Effects of Social Media on Self-Esteem and Narcissism

Research is beginning to explore the relationship between social media and personality traits, particularly positive self-views such as high self-esteem or higher narcissism. Although most Americans assume that self-esteem is the key to success, and low self-esteem a recipe for disaster, self-esteem actually has little effect on life outcomes (Baumeister, Campbell, Krueger, & Vohs, 2003). For example, Asian Americans have the lowest self-esteem of all U.S. ethnic groups (Twenge & Crocker, 2002), but perform the best academically. Narcissism, defined as an inflated sense of self, has some benefits (such as for public performance) but is not linked to long-term individual success and produces negative effects on others (for a review, see Twenge & Campbell, 2009).

Many correlational studies have found that narcissists have more friends on Facebook and are more active on the site (Bergman, Fearrington, Davenport, & Bergeman, 2011; Buffardi & Campbell, 2008; Carpenter, 2012; Gentile, Twenge, Freeman, & Campbell, 2012; Mehdizadeh, 2010; Rosen et al., 2012). Thus, the average person you are friends with on Facebook is more narcissistic than the average person you would encounter outside the site.

More recent experimental research has examined the direct causal relationship between social media use and self-views. In other words, do social media actually *cause* higher self-esteem and narcissism? Apparently, they do. College students who edited their MySpace page for 15 minutes (compared to those who traced a route on Google Maps for 15 minutes) scored higher on narcissistic personality traits. In a second study, using Facebook did not produce higher narcissism, but it did lead to higher self-esteem. Thus using social networking sites, even when editing one's page and not interacting with others, increases positive self-views (Gentile et al., 2012). Another study found that browsing Facebook (versus reading cnn.com) led to higher levels of narcissism and higher self-esteem among those who reported being particularly close to their Facebook friends. Facebook use also resulted in a focus on short-term pleasures such as spending money and eating unhealthy food (Wilcox & Stephen, in press).

Narcissism undermines beneficial social connections, as narcissism is correlated with decreased empathy and caring for others, in addition to increased aggression (Bushman et al., 2009; Campbell et al., 2007). Self-esteem does not usually cause aggression (Bushman et al., 2009), though some research suggests that, at least after an insult, higher self-esteem individuals are perceived as more abrasive (Heatherton & Vohs, 2000).

Overall, the available research suggests that spending time on social media such as Facebook leads to more focus on the self. And given that it sometimes increases narcissism as well, it possibly leads to less focus on others as well.

Generational Change in Caring for Others, Empathy, and Social Action

Indirect evidence of the effects of social media appears in studies of generational change. Just as people differ based on regional culture (e.g., the U.S. versus China), they differ based on the culture of the times (e.g., Baby Boomers raised in the 1950s versus GenMe raised in the 1990s; Twenge, 2006). GenMe, also known as Gen Y or Millennials, are the first generation to grow up with the Internet and were the first to join social networking sites such as MySpace and Facebook.

Of course, many cultural forces have influenced GenMe, including the increasing number of single-parent households, more cultural individualism, and growing racial and gender equality (Myers, 2000; Twenge, 2006). Yet GenMe's adoption of social networking sites and other technology is one of their most

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defining characteristics, especially compared to Generation X (born in the 1960s and 1970s) and Baby Boomers (born in the 1940s and 1950s).

Over the time when social media became prevalent, high school and college students declined in caring for others, empathy, and interest in social action and social issues. Between 1979 and 2008, college students' scores on the standard measure of interpersonal empathy for others declined significantly, especially after 2000 (Konrath, O'Brien, & Hsing, 2011). Over the same time, college students were increasingly likely to believe that people get what they deserve and to blame the victim (Malahy, Rubinlicht, & Kaiser, 2009).

The strongest evidence comes from nationally representative samples of high school and college students collected since the 1960s and 1970s. Compared to Boomers and GenX'ers, GenMe was significantly less likely to express empathy for outgroups, donate to charity, or say it was important to have a job worthwhile to society. Declines in civic orientation and social action were even more steep, with fewer saying they were interested in social problems of the nation and the world, fewer interested in government, and fewer interested in getting involved in politics (e.g., writing to a public official). Surprisingly, GenMe was also less likely than Boomers or GenX'ers to take action to help the environment and save energy (Twenge, Campbell, & Freeman, 2012). The only notable exception was an increase in community service; however, high schools increasingly required community service for graduation over this time period. Another in-depth, recent study of young adults concluded that many saw no reason to help others and that the vast majority (96%) were uninterested in civic and political affairs (Smith, Christoffersen, Davison, & Herzog, 2011).

These trends are reflected in the culture as a whole. In an analysis of American books since the 1800s in the Google Books database of 5 million books, the use of words concerned with morality and character declined over time (Kesebir & Kesebir, in press). In another Google Books analysis, the use of collectivistic pronouns such as *we* and *us* declined slightly between 1960 and 2008, and the use of individualistic pronouns such as *I*, *me*, and *you* increased (Twenge, Campbell, & Gentile, in press). In addition, individualistic words and phrases (*unique, self, I am special, I come first*) increased in use, whereas collectivistic words and phrases (*group, community, strength through unity, all in this together*) were unchanged or declined when controlled for individualistic language (Twenge, Campbell, & Gentile, 2012).

In addition, eleven studies showed evidence of increasing narcissism over the generations, including four datasets of college students collected over time between the 1980s and 2009 (Stewart & Bernhardt, 2010; Twenge & Foster, 2008, 2010; Twenge, Konrath, Campbell, Foster, & Bushman, 2008; for a review, see Twenge, in press).

These generational trends toward greater self-focus and away from social action were in place long before social media became prominent in the early 2000s. However, the era of social media has not caused a reversal of these trends.

Social Media and Recent Social Movements

Real-life examples of social action provide anecdotal evidence for the role of social media in civic engagement and protest movements. In the Middle East in 2011, social media helped fuel and organize the democratic uprisings known as the Arab Spring (Yousri, 2012). In these countries suffering under dictatorships, social media use by young people facilitated social and political action. In the United States, social media played a role in the 2011 Occupy protests, though their goals were not as clear.

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Social media also helped the viral spread of the Kony video—calling for the capture of war criminal Joseph Kony—in March 2012. The organization that produced the video quickly sold out of "action kits" with posters and stickers. The video engendered a backlash from some observers who noted that viewing a video and buying posters was *slacktivism*—a lazy version of protest (Bailyn, 2012). Sure enough, when the April 20 "day of action" named in the video arrived, few posters were used and even fewer people actually protested.

On the other hand, the video did succeed in raising awareness of Kony and his war crimes. Social media make it easier for people to donate money to causes and learn about global issues. Perhaps people are not getting directly involved themselves, but they can support worthy causes. The data from high school students suggest that social media are not leading to greater charity donation—charity donations have declined in this group since the 1990s. But perhaps charity donations will eventually increase as cell phones and social media make it easier to contribute small amounts.

We get the social movements we deserve. Social media are powerful tools for raising awareness and spreading the word for social action, political involvement, and protests. But if young citizens initially have little interest in these issues—and recent surveys suggest this is the case—then they are unlikely to use social media for these purposes.

Conclusions

Much research remains to be done on the effects of social media on caring for others and civic engagement. Existing research suggests that social media use has not led to more social action or empathy. Some evidence suggests it may, in fact, have led to less empathy, and in some cases to aggression against others. The exceptions are online political involvement (such as online petitions) and the perceived building of weak ties.

Social media, such as Facebook, allow students to expand their network of "friends," but may lead to them having fewer in-person friends (without the quotes)—those they can rely on for in-person emotional support and practical help (Turkle, 2012). Social media do boost positive self-views. This boost may have some benefits for self-esteem, but may lead to negative interactions with others if the boost also raises narcissism.

The usual caveat that most of these studies are correlational is not just a methodological note. More than likely, the same people who are addicted to drugs and alcohol are also likely to become addicted to Internet activity. Those same people may also lack the impulse control to inhibit aggression and help others. So those who spend many hours on the Internet may also be aggressive, antisocial, dislikeable people. The Internet probably did not make them this way, however. The helpful, likeable, mentally stable people are still online, but perhaps not for as many hours. Social media act as a prism, amplifying existing tendencies. When the existing tendencies of a culture are toward narcissism, civic disengagement, and individualism, that is the reality social media will reflect.

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